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Understanding Database Management

The volume of data shared across enterprises increases rapidly, making it difficult to manage their data. One possible solution for data management is Database Management Software (DBMS) which allows organizations to improve data accessibility and simplifies the process of documents management.

Database is an **organized collection of logically related structured information, or data** typically stored electronically in a computer system. Organisations databases can be stored on a file system, while large databases are hosted on computer clusters or cloud.

A **Database Management Software/System or DBMS** is an application used for storing, updating, retrieval, manipulating, and managing data stored in a database environment. It enables users to design a personalized database to meet their analytics and reporting needs. Database software design also supports creating, implementing, and maintaining an organization-wide data management system. Generally, most DBMS packages allow users to perform operations like database creation, storing data, updating data through SQL queries.

Main Features of a DBMS

An efficient database management system tool should include the following features:

1. **Data Normalization** - The risk of data duplication in a database is relatively high as multiple users share it simultaneously. Data normalization mitigates this risk and minimizes the chance of destructive anomalies appearing. No data redundancy and repetition save storage and significantly improve access time.
2. **User-defined rules and constraints** - Integrity constraints help organizations prevent accidental damage to the database by authorized users. A database management software allows users to define validation and integrity rules and conditions to ensure data satisfies the semantics.
3. **Security protocols** - Security controls protect the integrity of a database and the data and records residing in it. Some essential DBMS security controls include data encryption, user authentication, and user authorization.
4. **Data backup** - A backup protects your database against data loss. A copy of files stored in a database must be available to reconstruct data in case data get lost or corrupted. Most DBMS support logical and physical data backup.
5. **Data structuring** - A DBMS must allow users to organize information in a database in a clear hierarchical structure. It means all objects, records, and tables can be arranged systematically, like a catalog, so the records can easily be accessed and retrieved.

In addition, you must also look for various features and functionality depending on your use case. For example, a DBMS should offer easy database customization options, support multi-user access, and contain a metadata library.

Types of Database Management System

These can be broadly classified into four (4) types. The most popular types of database management systems with examples include:

- 1- **Hierarchical** - A Hierarchical Data management solution stores data in a parent-children relationship node, each representing a particular entity. This type of database management software allows one-to-one and one-to-many relationships, i.e., a parent node can have one or multiple child nodes, whereas the children node can only have one parent node.

Hierarchical database model

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An Object-Oriented DBMS — as the name suggests — is based on object-oriented programming (OOP). It's a data management solution type where entities are represented in objects and stored in memory. It provides a unified programming environment and is compatible with various programming languages, including Java, C++, .Net, and Visual Basic, to name a few.

Object 1: Maintenance Report					
Date					
Activity Code					
Route No.					
Daily Production					
Expenditure Hours					
Labor Hours					

Object 1 Instance	
01-12-01	
24	
195	
2.5	
6.0	
6.0	

Object 2: Maintenance Activity	
Activity Code	
Activity Name	
Production Unit	
Average Daily Production Rate	

Features and Functions of a database management system.

A DBMS such as Microsoft Access provides several functions to facilitate the efficient management and use of data. Some of the main functions of a DBMS are:

Data organization: A DBMS allows users to organize data into tables, with each table containing data about a specific topic or subject. Tables can be related to one another using keys and other constraints, allowing users to create complex relationships between different sets of data.

Data storage: A DBMS provides a way to store data in an efficient and organized manner. Data is stored in a structured format that allows for easy retrieval and manipulation.

Data retrieval: A DBMS allows users to retrieve data from a database quickly and easily. Queries can be created to extract data from one or more tables based on specific criteria, such as a date range or a customer ID.

Data security: A DBMS provides security features to ensure that data is protected from unauthorized access or modification. Users can be granted different levels of access to the database, with some users having read-only access while others have the ability to add, delete, or modify data.

Data backup and recovery: A DBMS provides features to backup and recover data in case of a system failure or other disaster. Backups can be scheduled automatically, and recovery options allow data to be restored to a previous state if necessary.

Data integrity: A DBMS provides tools to ensure that data is accurate and consistent. Constraints can be added to tables to enforce rules about the types of data that can be entered, preventing errors and ensuring that the data remains valid over time.

Data manipulation: A DBMS allows users to modify data in a database by adding, deleting, or updating records. Users can also create and modify tables, indexes, and other database objects.

Multi-user access: A DBMS allows multiple users to access the database simultaneously, ensuring that data remains consistent and up-to-date. This is achieved through transaction processing, which ensures that changes made by one user are isolated from changes made by others until they are committed to the database.

Indexing: A DBMS provides indexing features to speed up data retrieval by creating indexes on frequently used columns.

Data analysis: A DBMS provides tools to analyze data, such as report generation and data mining, to help users make informed decisions based on the data in the database.

What is Relational Database

A Relational database is a type of database that organizes data into one or more tables, or "relations". Each table consists of rows and columns, where each row represents a single record or instance of the data, and each column represents a specific attribute or characteristic of that data.

The **relationships** between tables are established by creating common fields between them, known as keys, which enable the linking of data across multiple tables. These relationships allow the data to be organized logically and efficiently, making it easier to search, retrieve, and update.

Relational database model provides a highly structured, flexible, and scalable approach to storing and managing data, making it an ideal choice for many business and enterprise applications.

The **relational database model** was first introduced by E.F. Codd in the 1970s and has since become the dominant database model used in commercial and enterprise applications. The most commonly used relational database management systems (RDBMS) are MySQL, Oracle, Microsoft SQL Server, and PostgreSQL.

Relationships are connections between tables that allow data to be organized and retrieved in a meaningful way. There are three types of relationships: one-to-one, one-to-many, and many-to-many. One-to-one relationships occur when each row in one table corresponds to one and only one row in another table. One-to-many relationships occur when each row in one table corresponds to one or more rows in another table. Many-to-many relationships occur when each row in one table corresponds to one or more rows in another table, and vice versa.

Some of the **Advantages** of using a relational database include:

Data Integrity: The relational database model enforces rules to maintain data integrity, ensuring that each piece of data is accurate, consistent, and up-to-date.

Scalability: The ability to add or modify tables and fields within a relational database without affecting existing data makes it highly scalable and adaptable to changing business needs.

Flexibility: Relational databases are highly flexible and can be used to store a wide variety of data types, including text, numbers, images, and multimedia.

Security: Relational databases offer a high degree of security through features such as user authentication, access control, and encryption.

Data Consistency: Since data is stored in a centralized location, it is easier to maintain data consistency across multiple users and applications.

Relational Database (RDB) - This is a kind of database which contains a collection of more than one tables of data from which data can be accessed in different ways without reorganizing the tables all of which are formally organised according to a relational model. A RDB has the ability to establish links—or relationships—between information by joining tables, which makes it easy to understand and gain insights about the relationship between various data points.

- Is a collection of information that organizes data with defined relationships for easy access.
- It is a collection of data items with pre-defined relationships between them.
- It is a type of database that focuses on the relation between stored data elements.
- The data from the Relational database can be accessed, modified or managed by querying with the SQL (structured query language) programming language.

Relational database management systems (RDBMS)

This is the most common type of DBMS. They are used to interact with databases that contain structured data in a table format with predefined relationships. Moreover, they use structured query language (SQL) to interact with databases. Microsoft SQL Server, MySQL, PostgreSQL, MariaDB and Oracle Database are some popular DBMS that come under this category. A relational database must have more than one table. An Access desktop database (.accdb or .mdb) is a fully functional RDBMS.

Applications of Database Management Systems

The application areas of a DBMS are vast and varied. Some of the key application areas of a DBMS include:

- **Business:** DBMS is used extensively in the business world to store and manage large amounts of data related to customers, products, transactions, sales, inventory, and employees. It is used to run business operations smoothly, improve decision-making, and increase productivity.
- **Sales and marketing** - Prospect and customer information is stored and accessed via databases.
- **Banks** - In the banking sector, DBMS is used to store client info, account activities, disbursements, credits, and mortgages.
- **Economics and Finance:** DBMS is used to store and manage financial data such as transactions, accounts, and investments. It helps financial institutions to process transactions faster, detect fraud, and manage risk effectively. DBMS stores data about bonds, transactions, and acquisitions of fiscal instruments, such as shares and stocks
- **Healthcare:** DBMS is used to store and manage medical data such as patient records, medical histories, test results, and prescriptions. It helps healthcare providers to provide better patient care, manage resources effectively, and comply with regulations.
- **Education:** DBMS is used to store and manage educational data such as student registration records, course schedules, and grades and results. It helps educational institutions to track student progress, manage resources, and analyze data for continuous improvement.
- **Government:** DBMS is used to store and manage government data such as census data, tax records, and public records. It helps governments to provide better services to citizens, manage resources effectively, and comply with regulations.
- **Research:** DBMS is used to store and manage research data such as experimental results, scientific papers, and patents. It helps researchers to collaborate, manage data effectively, and analyze data for insights.
- **E-commerce:** DBMS is used to store and manage data related to online shopping such as product catalogs, customer data, and sales data. It helps e-commerce businesses to provide personalized shopping experiences, improve customer retention, and optimize operations.
- **Airlines:** Flight bookings and scheduling info is stored in databases.
- **Telecommunication:** Databases store call archives, monthly bills, retaining balances, and other call-related information.
- **Human Resources:** Records about workers, remunerations, payroll, deduction, generating salaries, and more information are kept in database systems.

Data type in a Database

In a database, there are several types of data types used to define the type of data that can be stored in each field of a table. The most common data types include:

Integer: This data type is used to store whole numbers without any decimal places. It can be used to represent numbers such as age, quantity, or any other numeric value that does not require decimal places.

Float/Double: This data type is used to store floating-point numbers with decimal places. It is commonly used to store values such as prices, percentages, or any other numeric value that requires decimal places. FLOAT value is approximate which means not all values can be represented exactly.

Boolean: This data type is used to store a true/false value. It is commonly used to represent binary values such as Yes/No or True/False.

Character/String: This data type is used to store text data. It can store any combination of letters, numbers, and symbols. It is commonly used to store data such as names, addresses, and descriptions.

Date/Time: This data type is used to store date and time values. It is commonly used to store data such as birth dates, transaction dates, or any other date/time-related information.

Blob: This data type is used to store large binary data such as images, videos, or any other type of file.

Decimal: This data type is used to store fixed-point (precision) numbers with decimal places. It is commonly used to store values such as currency or any other numeric value that requires a fixed number of decimal places. Eg, 12.58, 11.23, 10.03

The choice of data type for a field depends on the type of data that will be stored in it, as well as the requirements of the application or system that will be using the database.

Other Data types Used in Ms Access

Every field in a table has properties and these properties define the field's characteristics and behaviour. The most important property for a field is its data type. A field's data type determines what kind of data it can store. MS Access supports different types of data, each with a specific purpose.

- The data type determines the kind of the values that users can store in any given field.
- Each field can store data consisting of only a single data type.

Here are some of the most common data types you will find used in a typical Microsoft Access database.

Type of Data	Description	Size
Short Text	Text or combinations of text and numbers, including numbers that do not require calculating (e.g. phone numbers).	Up to 255 characters.

Long Text	Lengthy text or combinations of text and numbers.	Up to 63, 999 characters.
Number	Numeric data used in mathematical calculations.	1, 2, 4, or 8 bytes (16 bytes if set to Replication ID).
Date/Time	Date and time values for the years 100 through 9999.	8 bytes
Currency	Currency values and numeric data used in mathematical calculations involving data with one to four decimal places.	8 bytes
AutoNumber	A unique sequential (incremented by 1) number or random number assigned by Microsoft Access whenever a new record is added to a table.	4 bytes (16 bytes if set to Replication ID).
Yes/No	Yes and No values and fields that contain only one of two values (Yes/No, True/False, or On/Off).	1 bit.

Here are some of the other **more specialized** data types, you can choose from in Ms Access.

Data Types	Description	Size
Attachment	Files, such as digital photos. Multiple files can be attached per record. This data type is not available in earlier versions of Access.	Up to about 2 GB.
OLE objects	OLE objects can store pictures, audio, video, or other BLOBs (Binary Large Objects)	Up to about 2 GB.
Hyperlink	Text or combinations of text and numbers stored as text and used as a hyperlink address.	Up to 8,192 (each part of a Hyperlink data type can contain up to 2048 characters).
Lookup Wizard	A simple lookup field uses the contents of another table or a value list to validate the contents of a single value per row. A complex lookup field allows you to store multiple values of the same data type in each row. It is found in design view	Dependent on the data type of the lookup field.

Calculated

You can create an expression that uses data from one or more fields. You can designate different result data types from the expression.

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Database Objects

1. **Table** – A table is a database object that stores data in an organised form, it consists of fields and records. In Access, all data is stored in **tables**, which puts tables at the heart of any database. It is a basic unit of storage. Each row is a **record** and each column is a **field** within each record. A record is all of the data pertaining to a person, place, thing or event. Each record will have one or more fields and every database must have at least one table.

Customers						
CustomerID		First Name	Last Name	Phone	Street Address	City
						State/Province
						Zip/Postal Code
						Click to Add
1	Devinder	Patel	218-555-5641	212 Ash Street	Aurora	MN
						59990
2	Maria	Escobar	218-555-5146	212 Birch Street	Lake Cosmos	MN
						59991
3	Ahmed	Rashed	218-555-1656	212 Cedar Street	Park Falls	MN
						59992
4	Arnold	Hammer	218-555-8811	212 Elm Street	Aurora	MN
						59990
5	Susan	Wong	(218) 555-0182	212 Fir Street	Aurora	MN
						59990
* (New)						

Components of a Database table.

Record: A collection of data about in individual item on a table

Field: A single item of Data common to all records on a table

Tuple: A group of two or more records in a table.

Schema: A group of two or more tables

Primary Key: It is a field in a table that uniquely identifies each record on a database table. A primary key must contain unique values. Databases use the key to compare, sort, and store records and also create relationships between records. A primary key is used to ensure data in the specific column is unique.

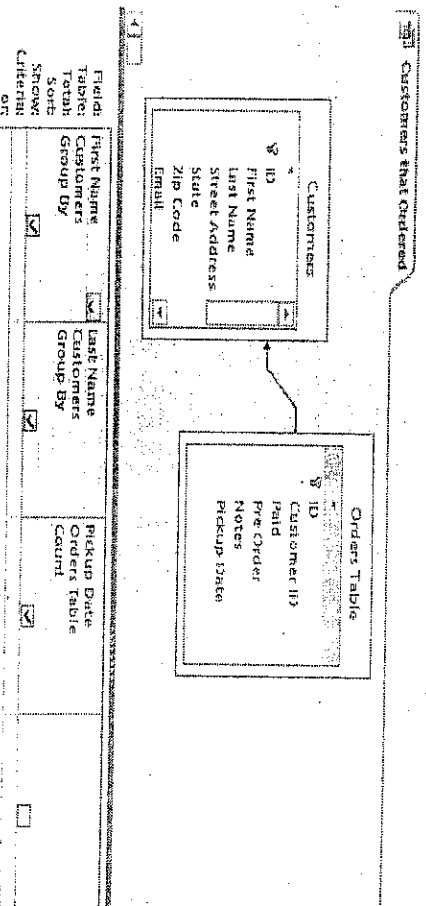
- It may consist of a single attribute or multiple attributes in combination.
- It must be entered when a record is created and never to be changed.
- Each table in a database has a column or two as primary key.
- Only one primary key is allowed in a table and it doesn't allow NULL values.
- Its value cannot be deleted from the parent table.

Foreign Key: It is a field in a relational table that matches the primary key field of another table. It is used to link two tables together.

- more than one foreign key are/can be allowed in a table.
 - It can also contain NULL values.
 - Its value can be deleted from the child table.
2. **Query** – A database query is a way of requesting information from the database or from several tables, a query most common function is to retrieve specific data from tables. The data users want to see is usually spread across several tables, queries allow you to view it in a single datasheet.

The Function of a query is to retrieve data that meets specific conditions from tables. You can also use it to perform some predefined calculations. On the Access desktop application, you can use queries for updating or deleting multiple records simultaneously. Queries can be of two types: **select queries** and **action queries**. While a select query allows users to view, copy, and print retrieved data, an action query can perform a task with the data.

- Running a query is like asking a detailed question of your database.
- When you build a query in Access, you are defining specific search conditions to find exactly the data you want.
- In Access, you can use the graphical query by example facility or you can write Structured Query Language (SQL) statements to create your queries.
- You can define queries to Select, Update, Insert, or Delete data.
- You can also define queries that create new tables from data in one or more existing tables.



3. **Forms** - is an object in a database designed primarily for data entry/input or display or for control of application execution. You use forms to customize the presentation of data that your application extracts from queries or tables. Forms provide an easy way
 - Forms are used for entering, modifying, and viewing records.
 - The reason forms are used so often is that they are an easy way to guide people toward entering data correctly.
 - When you enter information into a form in Ms Access, the data goes exactly where the database designer wants it to go in one or more related tables.

Customers

Search

First Name: [] Last Name: []

Street Address: [7 East Walker Dr.] City: [Raleigh] State: [NC] Zip Code: [27612]

Email: [hacker@unmail.com] Phone Number: [919-555-2314]

[New Record] [Back] [Find]

Record: 1 of 200

4. **Reports** - is an object in databases designed for Displaying, formatting, Viewing, calculating, printing, presenting and summarizing selected data in a table or group of tables. You can choose where the info appears on the printed pages.
- Reports will organize and group the information in a table or query and provide a way to print the data in a database.

Reports are database objects that can be used to present information in your database for any of the following purposes: To display or distribute a summary of the data in your database, snapshots of the data to be archived. Details about individual records should be provided.

- You can view a report on your screen before you print it.
- If forms are for input purposes, then reports are for output.
- Anything you plan to print deserves a report, whether it is a list of names and addresses, a financial summary for a period, or a set of mailing labels.
- Reports are useful because they allow you to present components of your database in an easy-to-read format.
- You can even customize a report's appearance to make it visually appealing.
- Ms Access offers you the ability to create a report from any table or query.

5. Macro

A macro is a tool that automates tasks and adds functionality to various database parts, including forms, reports, and controls. It's a program to automate manual tasks like opening a report, running a query, copy-pasting, and table formatting in Access that you do regularly. This object is a structured definition of one or more actions that you want Access to perform in response to a defined event. An Access Macro is a script for doing some job. For example, to create a button which opens a report, you could use a macro which will fire Open Report action.

- You can include simple conditions in macros to specify when one or more actions in the macro should be performed or skipped.
- You can use macros to open and execute queries, to open tables, or to print or view reports.
- You can also run other macros or Visual Basic procedures from within a macro.
- Data macros can be attached directly to table events such as inserting new records, editing existing records, or deleting records.
- Data macros in web apps can also be stand-alone objects that can be called from other data macros or macro objects.

6. Module

Module is an object in desktop databases containing custom procedures that you code using Visual Basic. Modules provide a more discrete flow of actions and allow you to trap errors.

- Everything that can be done in a macro can also be done in a module, but you don't get the macro interface that prompts you what is needed for each action.
- Modules are far more powerful, and are essential if you plan to write code for a multi-user environment, because macros cannot include error handling.
- Modules can be standalone objects containing functions that can be called from anywhere in your application, or they can be directly associated with a form or a report to respond to events on the associated form or report.

Important Concepts in Databases Management.

Normalization - is a database design technique that reduces data redundancy and eliminates undesirable characteristics like Insertion, Update and Deletion Anomalies. Normalization rules divides larger tables into smaller tables and links them using relationships. The purpose of Normalisation in database is to eliminate redundant (repetitive) data and ensure data is stored logically.

Database Administrator — Is an Expert responsible for the design and management of an organization's databases as well as the evaluation, selection and implementation of the database management system (DBMS) software. He/She;

- Maintain and protect sensitive information and provide access to important datasets in the organisation.
- Back up, restore, and troubleshoot database, sets system access, updating and integrating old programs to implement the latest technology

Data Administrator - Data administrator is a person who is responsible for processing data into a convenient data model. The person is in charge of figuring out which data is relevant to be stored in the database. Data Administrator is less of a technical role and more of a business role with some technical knowledge. This role is also known as Data Analyst. So, it is mostly a high level function which is responsible for the overall management of data resources in an organization.

Database Model — A database model shows the logical structure of a database, including the relationships and constraints that determine how data can be stored and accessed. A data model is a structure of the data that contains all the required details of the data like the name of the data, size of the data, relationship with other data and constraints that are applied on the data.

Field Validation Rule: If you set a validation rule for a field, Access prevents you from leaving that field unless you enter a valid value. Suppose you add >= #12/01/2021# in the property of Validation Rule of the date field. In that case, you need to enter dates only on and after 1 December 2021.

Record Validation Rule: You can use it to control when you can save a record. This rule considers other fields in the same table and compares their values.